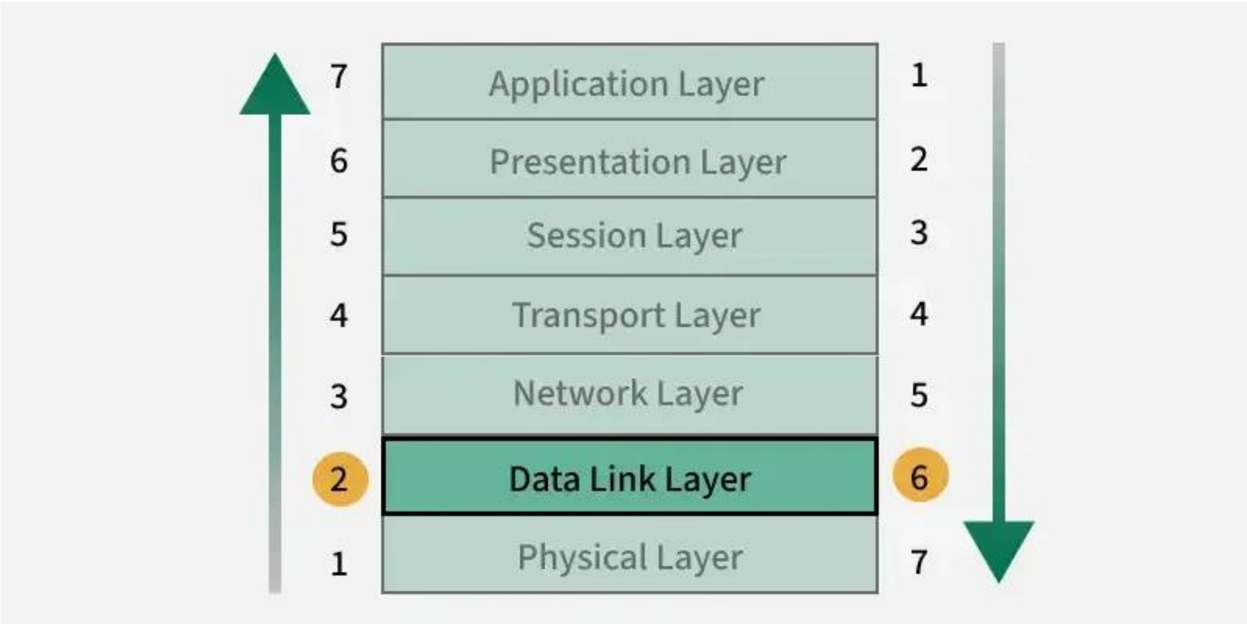
Data Link Layer in OSI Model

The data link layer is the second layer from the bottom in the [OSI](OSI)(Open System Interconnection) network architecture model. It is responsible for the node-to-node delivery of data within the same local network. Its major role is to ensure error-free transmission of information. DLL is also responsible for encoding, decoding, and organizing the outgoing and incoming data.

This is considered the most complex layer of the OSI model as it hides all the underlying complexities of the hardware from the other above layers. In this article, we will discuss Data Link Layer in Detail along with its functions, and sub-layers.



Data Link Layer in OSI Model

Sub-Layers of The Data Link Layer

The data link layer is further divided into two sub-layers, which are as follows:

Logical Link Control (LLC)

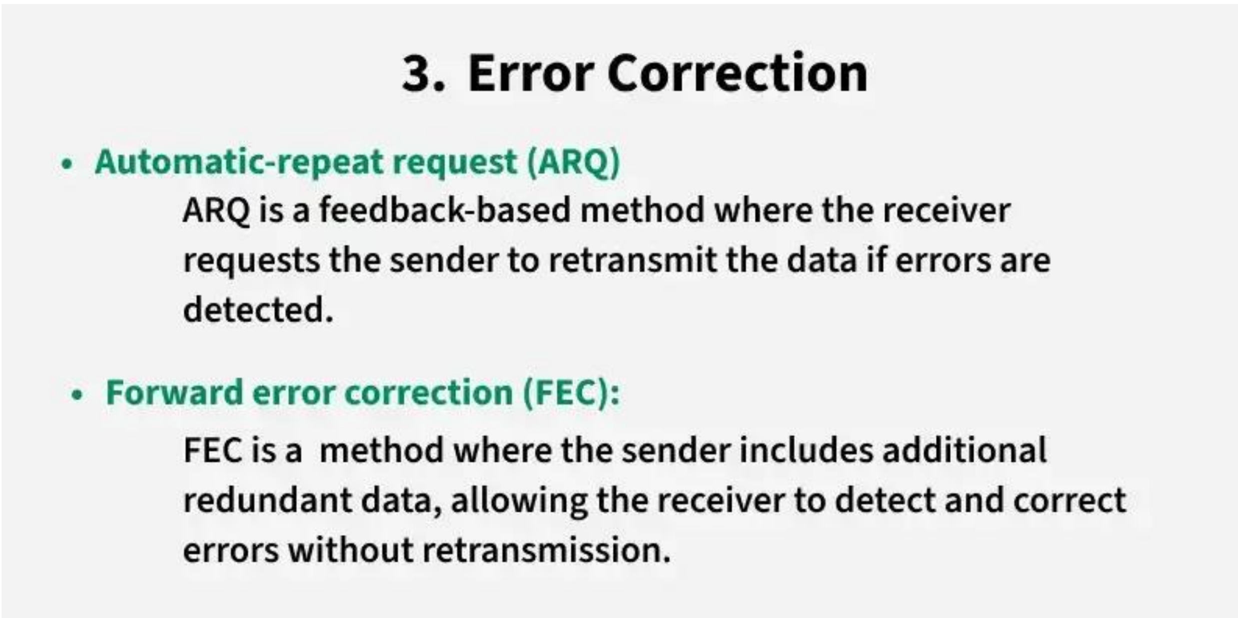
This sublayer of the data link layer deals with multiplexing, the flow of data among applications and other services, and LLC is responsible for providing error messages and acknowledgments as well.

Media Access Control (MAC)

MAC sublayer manages the device's interaction, responsible for addressing frames, and also controls physical media access. The data link layer receives the information in the form of packets from the Network layer, it divides packets into frames and sends those frames bit-by-bit to the underlying physical layer.

Functions of The Data-link Layer

There are various benefits of data link layers s let's look into it.



The primary functions of the DLL are:

- **Framing: Organizing raw data into manageable frames.**
- **Flow Control: Regulating the rate of data transfer to prevent congestion.**
- **Error Control: Ensuring data is received correctly without errors.**
- **Link Management: Handling how devices access the shared medium (like Ethernet or Wi-Fi).**

Nodes and Links

- **Nodes: A node is any device in the network that can send or receive data. Examples include routers, switches, computers, and mobile devices.**
- **Links: Links are the physical or logical connections between two nodes. These can be wired (e.g., Ethernet cables) or wireless (e.g., Wi-Fi, Bluetooth).**

Node-to-node communication happens at the Data Link Layer, which ensures that the data sent from one node reaches another node over the link without errors.

Protocols in Data link layer

There are various protocols in the data link layer, which are as follows:

- Synchronous Data Link Protocol (SDLC)
- High-Level Data Link Protocol (HDLC)
- Serial Line Interface Protocol (SLIP)
- Point to Point Protocol (PPP)
- Link Access Procedure (LAP)
- Link Control Protocol (LCP)
- Network Control Protocol (NCP)

Devices Operating at the Data Link Layer

1.Switch

- A switch is a key device in the Data Link Layer.
- It uses **MAC addresses** to forward data frames to the correct device within a network.
- Works in **local area networks (LANs)** to connect multiple devices.

2. Bridge

- A bridge connects two or more LANs, creating a single, unified network.
- Operates at the Data Link Layer by forwarding frames based on **MAC addresses**.
- Used to reduce network traffic and segment a network.

3. Network Interface Card (NIC)

- A NIC is a hardware component in devices like computers and printers.
- Responsible for adding the **MAC address** to frames and ensuring proper communication with the network.
- Operates at the Data Link Layer by preparing and sending frames over the physical medium.

4. Wireless Access Point (WAP)

- A WAP allows wireless devices to connect to a wired network.
- Operates at the Data Link Layer by managing **wireless MAC addresses**.

- Uses protocols like Wi-Fi (IEEE 802.11) to communicate with devices.

5. Layer 2 Switches

- These are specialized switches that only operate at Layer 2, unlike multi-layer switches.
- Responsible for **frame forwarding** using MAC address tables.

Limitations of Data Link Layer

- **Limited Scope**: It operates only within a local network and cannot handle end-to-end communication across different networks.
- **Increased Overhead**: Adding headers, trailers, and redundant data (for error correction) increases the size of transmitted data.
- **Error Handling Dependency**: While it can detect and correct some errors, it relies on upper layers for handling more complex issues.
- **No Routing Capability**: The Data Link Layer cannot make routing decisions. It only ensures delivery within the same network segment.
- **Resource Usage**: Flow control and error correction mechanisms may consume extra processing power and memory

Applications of Data Link Layer

- **Local Area Networks (LANs)**: Enables reliable communication between devices within a local network using protocols like Ethernet (IEEE 802.3).
- **Wireless Networks (Wi-Fi)**: Manages communication between devices in wireless networks via protocols like IEEE 802.11 hence, handling media access and error control.
- **Switches and MAC Addressing**: Facilitates the operation of switches by using MAC addresses to forward data frames to the correct device within the network.
- **Point-to-Point Connections**: Used in protocols like PPP (Point-to-Point Protocol) to establish and manage direct communication between two nodes.

Services of the Data Link Layer

The services provided by the Data Link Layer can be categorized as:

1. **Framing**:

- The DLL divides data into frames to prepare for transmission. A frame consists of a header (with control information like address and error detection) and a payload (the actual data being sent).

2. **Flow Control**:
   - Flow control regulates the data rate between sender and receiver to ensure the receiver is not overwhelmed. Techniques like **Stop-and-Wait** and **Sliding Window** are commonly used for this.
3. **Error Control**:
   - DLL ensures reliable data transmission by detecting and correcting errors. It uses mechanisms like **Parity Bits**, **Checksums**, and **Cyclic Redundancy Check (CRC)** to identify errors in frames. If errors are detected, the frame may be retransmitted.
4. **Access Control**:
   - DLL manages how nodes access a shared medium. For example, **Ethernet** uses **Carrier Sense Multiple Access with Collision Detection (CSMA/CD)** to control access to the medium in a shared network.

Framing

Framing is the process of dividing data into smaller, manageable units called **frames** for transmission over the physical medium. Frames typically contain:

- **Header**: Information like the source and destination addresses, type of data, and other control information.
- **Payload**: The actual data being transmitted.
- **Trailer**: Includes error-checking information such as CRC (Cyclic Redundancy Check).

Frames help the receiving node determine where one piece of data ends and another begins, ensuring correct interpretation of the transmitted bits.

Flow Control

Flow control mechanisms regulate the data flow between the sender and receiver to ensure the receiver is not overwhelmed with too much data at once. Common flow control techniques are:

1. **Stop-and-Wait**: The sender waits for an acknowledgment after sending each frame before sending the next one. Simple but inefficient for high-speed networks.

2. **Sliding Window**: This allows the sender to transmit several frames before needing an acknowledgment. The window size controls how many frames can be sent without acknowledgment.

Error Control

Error control ensures that the data transmitted over the network is accurate and reliable. It involves two main processes:

1. **Error Detection**: Methods like **Parity Check**, **Checksums**, and **Cyclic Redundancy Check (CRC)** are used to detect errors in data frames.
2. **Error Correction**: If an error is detected, the frame may be discarded and retransmitted. In more advanced error correction schemes (like **Forward Error Correction (FEC)**), the receiver can automatically correct certain types of errors.

Congestion Control

Congestion control ensures that the network does not become overloaded, which can cause performance degradation. While congestion control is typically associated with higher layers (like the transport layer), the Data Link Layer also plays a role in ensuring data is not sent faster than it can be processed.

Techniques include:

- **Backpressure**: If the receiving node cannot handle more data, it sends a signal to the sender to stop or slow down the data transmission.
- **Window Size Adjustment**: Dynamic adjustment of the sliding window in flow control can also help prevent congestion by limiting the number of frames in flight at once.

Data Link Addressing

Data Link Layer addressing is used to identify devices within the same network segment (or link). It is essential for communication between devices on the same local area network (LAN). The address used in the DLL is called the **MAC (Media Access Control) address**.

- **MAC Address**: A unique identifier assigned to each network interface card (NIC) by the manufacturer. It operates at the DLL and is used to distinguish devices on a local network.

- **Ethernet Frames**: Ethernet uses MAC addresses to identify the source and destination nodes in a frame.

ARP (Address Resolution Protocol)

ARP is a protocol used to map an **IP address** (network layer address) to a **MAC address** (data link layer address) within a local network. It allows devices to determine the MAC address of a device when only its IP address is known.

The process works as follows:

1. A device sends an **ARP request** to the local network, asking, "Who has IP address X? Tell me your MAC address."
2. The device with the matching IP address replies with an **ARP reply**, providing its MAC address.
3. The requesting device then stores this MAC address in its **ARP cache** for future use.

ARP is essential for devices to communicate at the data link layer using MAC addresses after identifying the destination's IP address.

Communication Using the Data Link Layer

1. **Point-to-Point Communication**:
   - In a **point-to-point link**, such as between two devices connected directly (like through a cable), the DLL ensures error-free communication, manages flow control, and addresses the devices using MAC addresses.
2. **Broadcast Communication**:
   - In a **broadcast network** (such as Ethernet), the DLL is responsible for broadcasting frames to all devices on the network and determining which device should process the frame based on its MAC address.
3. **Link Layer Protocols**:
   - Examples of protocols operating at the Data Link Layer include **Ethernet**, **Wi-Fi (IEEE 802.11)**, **PPP (Point-to-Point Protocol)**, and **HDLC (High-Level Data Link Control)**. These protocols define how data is framed, addressed, and transmitted over various types of physical links.